

Homework Set HW10 due 4/1/04 at 11:00 PM

You may need to give 4 or 5 significant digits for some (floating point) numerical answers in order to have them accepted by the computer.

1.(1 pt)

This is problem 3 Section 4.5 page 236.

Evaluate the limit

$$\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1} = \underline{\hspace{2cm}}$$

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF*, if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

2.(1 pt)

This is problem 6 Section 4.5 page 236.

Evaluate the limit

$$\lim_{x \rightarrow -1} \frac{x^{10} - 1}{x + 1} = \underline{\hspace{2cm}}$$

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF*, if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

3.(1 pt)

This is problem 8 Section 4.5 page 236.

Evaluate the limit

$$\lim_{x \rightarrow 0} \frac{(1 - \cos^2(x))}{3 \sin(x)} = \underline{\hspace{2cm}}$$

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF*, if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

4.(1 pt)

This is problem 10 Section 4.5 page 236.

Evaluate the limit

$$\lim_{x \rightarrow 0} \frac{(1 - \cos(x))}{x^2} = \underline{\hspace{2cm}}$$

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF*, if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

5.(1 pt)

This is problem 12 Section 4.5 page 236.

Evaluate the limit

$$\lim_{x \rightarrow 0} \frac{\tan(3x)}{\sin(5x)} = \underline{\hspace{2cm}}$$

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF*, if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

6.(1 pt)

This is problem 18 Section 4.5 page 236.

Evaluate the limit

$$\lim_{x \rightarrow \pi/2} \frac{\sin(2x) \cos(x)}{x \sin(4x)} = \underline{\hspace{2cm}}$$

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF*, if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

7.(1 pt)

This is problem 20 Section 4.5 page 236.

Evaluate the limit

$$\lim_{x \rightarrow 0} x^2 \sin\left(\frac{1}{x}\right) = \underline{\hspace{2cm}}$$

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF*, if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

8.(1 pt)

This is problem 24 Section 4.5 page 236.

Evaluate the limit

$$\lim_{\theta \rightarrow 0} \frac{\theta - 1 + \cos^2(\theta)}{\theta^2 + 5\theta} = \underline{\hspace{2cm}}$$

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF*, if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

9.(1 pt)

This is problem 28 Section 4.5 page 236.

Evaluate the limit

$$\lim_{x \rightarrow +\infty} x^{-5} \ln(x) = \underline{\hspace{2cm}}$$

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF*, if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

10.(1 pt)

This is problem 30 Section 4.5 page 236.

Evaluate the limit

$$\lim_{x \rightarrow 0^+} \sin(x) \ln(x) = \underline{\hspace{2cm}}$$

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF*, if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

11.(1 pt)

This is problem 32 Section 4.5 page 236.

Evaluate the limit

$$\lim_{x \rightarrow -\infty} \left(1 - \frac{3}{x}\right)^{2x} = \underline{\hspace{2cm}}$$

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF*, if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

12.(1 pt)

This is problem 38 Section 4.5 page 236.

Evaluate the limit

$$\lim_{x \rightarrow +\infty} (\sqrt{x^2 - x} - x) = \underline{\hspace{2cm}}$$

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF*, if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

13.(1 pt)

This is problem 44 Section 4.5 page 236.

Evaluate the limit

$$\lim_{x \rightarrow 0^+} (e^x - 1)^{1/\ln(x)} = \underline{\hspace{2cm}}$$

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF* , if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

14.(1 pt)

This is problem 47 Section 4.5 page 236.

Evaluate the limit

$$\lim_{x \rightarrow 0^+} \left(\frac{2 \cos(x)}{\sin(2x)} - \frac{1}{x} \right) = \underline{\hspace{2cm}}$$

In this exercise, if the limit does not exist but is $+\infty$ then enter *INF* , if it is $-\infty$ enter *-INF* and if the limit does not exist and is neither of these two cases enter *DNE*

15.(1 pt)

This is problem 7 Section 4.6 page 245.

Find the rectangle of largest area that can be inscribed in a semicircle of radius R , assuming that one side of the rectangle lies on the diameter of the semicircle.

NOTE: Let L denote the length of the side that lies on the diameter and H denote the height of the rectangle. Your answer will likely involve R .

$$L = \underline{\hspace{2cm}}$$

$$H = \underline{\hspace{2cm}}$$

16.(1 pt)

This is problem 13 Section 4.6 page 246.

A truck is 250 *mi* due east of a sports car and is traveling west at a constant speed of 60 *mph* . Meanwhile, the sports car is going north at 80 *mph* . When will the truck and the car be closest to each other? What is the minimum distance between them.

HINT: First minimize the square of the distance between them.

$$\text{Time they are closest } t = \underline{\hspace{2cm}} \text{ } hr$$

$$\text{Minimum distance } D = \underline{\hspace{2cm}} \text{ } mi$$

17.(1 pt)

This is problem 16 Section 4.6 page 246.

A closed box with square base is to be built to house an ant colony. The bottom of the box and all four sides are to be made of material costing 1 dollar/sq ft, and the top is to be constructed of glass costing 5 dollar/sq ft. What are the dimensions of the box of greatest volume that can be constructed for 72 dollars?

NOTE: Let x denote the length of the side of the base and y denote the height of the box.

$$\text{side of base } x = \underline{\hspace{2cm}} \text{ } ft$$

$$\text{height } y = \underline{\hspace{2cm}} \text{ } ft$$

18.(1 pt)

This is problem 18 Section 4.6 page 246.

The bottom of an 8-foot-high mural painted on a vertical wall is 13 feet above the ground. The lens of a camera fixed to a tripod is 4 feet above the ground. How far from the wall should the camera be placed to photograph the mural with the largest possible angle?

NOTE: Let x denote the horizontal distance from the wall to the camera. Refer to the figure in the book on page 246.

$$x = \underline{\hspace{2cm}} \text{ } ft$$

19.(1 pt)

This is problem 20 Section 4.6 page 246.

Two towns A and B are 12 miles apart and are located 5 and 3 miles, respectively, from a long, straight highway (see figure 4.62 page 246).

A construction company has a contract to build a road from A to the highway and then to B. Analyze a model to determine the length (to the nearest tenth of a mile) of the *shortest* road that meets these requirements.

$$\text{Minimum distance: } \underline{\hspace{2cm}} \text{ } mi$$

20.(1 pt)

This is problem 21 Section 4.6 page 246.

A poster is to contain 108 cm^2 of printed matter, with margins of 6 cm at the top and bottom and 2 cm on the sides. What is the minimum cost of the poster if it is to be made of material costing 20 cents/ cm^2 ?

NOTE: Draw a rectangle figure and let x denote the horizontal dimension of the printed matter and y the vertical dimension. Find a formula involving x and y for the area of the poster. You need to minimize the area. Then write a formula involving x and y for the cost C of material for the poster.

$$x = \underline{\hspace{2cm}} \text{ } in$$

$$y = \underline{\hspace{2cm}} \text{ } in$$

$$C = \underline{\hspace{2cm}} \text{ } dollars$$

21.(1 pt)

This is problem 24 Section 4.6 page 247.

A storage bin is to be constructed by removing a sector with angle θ from a circular piece of tin of radius 10 feet and folding the remainder of the tin to form a cone (see figure 4.64 on page 247). What is the maximum volume of a storage bin formed in this fashion?

$$\text{angle } \theta: \underline{\hspace{2cm}} \text{ } radians$$

$$\text{Maximum volume (to the nearest tenth): } \underline{\hspace{2cm}} \text{ } cubic feet$$